

Eric Miao

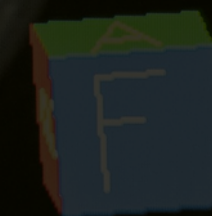
EEG-Based Detection of Cognitive Overload in Students Using an Engagement Index

PHA

MF 1.00

TR 6690.0
TE 93.0
BW 100.0
-2 MPB/MPB

Tra>Cor -13
>Sag -4



TP H17

Viewing

Filming

3D

Why

“Time to Take a Break”

Parents and teachers often prompt students to take breaks. Many students use studying techniques such as the Pomodoro, which instructs students to cycle through 25 minutes of work and 5 minutes of rest.

But every student is different and can focus for varying amounts of time. Also, what if a student has a succession of many cognitively intensive classes?

The reality is that a one-size-fits-all structure assumes that attention is uniform and predictable. It isn't. A static break structure may help some students, but for others, it may come too late or cut them off way too early.

If we want to optimize how we learn in the future, we need a system that adapts to the individuals cognitive states instead of a model that forces everyone into the same timing model.

Neuroplasticity

Neuroplasticity is defined as the brain's ability to reorganize and strengthen neural connections in response to experiences, environment and injury. This means that our brain can actually adapt and strengthen just like how our muscles do. In the gym, this might look like choosing weights that are challenging enough to push you, but not so heavy that your form breaks down. Our brain is no different.

By creating a system that tracks our brain engagement in real time, we can detect when “cognitive form” begins to decline or when cognitive load is accumulating even before performance drops, signalling that a break is needed. Early intervention prevents full disengagement and protects mental resources. A closed-loop system can then deliver timely breaks based on neural data, allowing longer and more effective focus sessions.

Evolution of technologies

Over the course of a couple of decades, humanity has made major strides towards understanding the brain, creating an assortment of brain computer interfaces (BCI) technologies.

As primitive EEG-based BCIs began to emerge in the early 1900s, researchers managed to measure electrical activity from the scalp externally, attempting to bridge these patterns to certain mental states. Unfortunately, these systems were large, relying on bulky machines and controlled lab settings, making them impossible to scale.

Today, technology looks very different. Sensors are smaller. Devices are portable. Consumer-grade BCIs can be worn like a headband or a pair of glasses. It can now be tested in real-world settings — like classrooms — which makes a study like this possible.

Research Question

How can EEG signals gathered from consumer-grade wearables be processed to reliably detect cognitive engagement in high school students during sustained academic tasks?

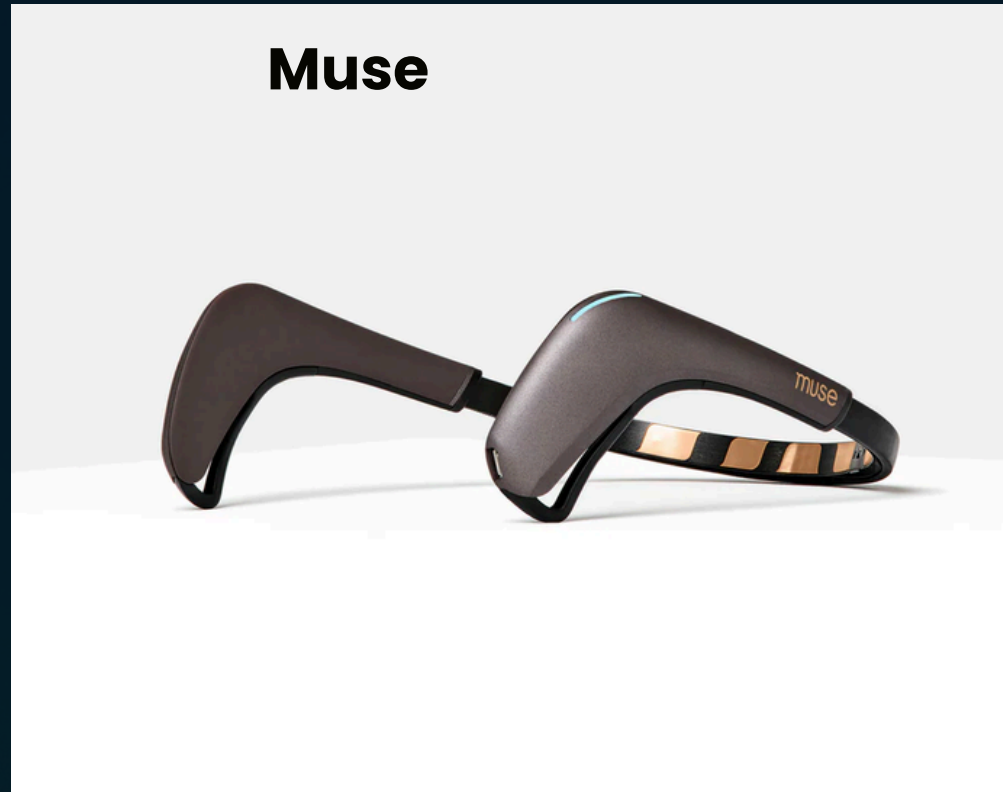
Hypothesis

If cognitive engagement decreases during sustained studying, then the Engagement Index derived from EEG signals will significantly decrease relative to baseline levels.

Status Quo

- Only measures frontal lobe
- Sleek formfactor
- 4 electrodes
- Cheaper

Muse

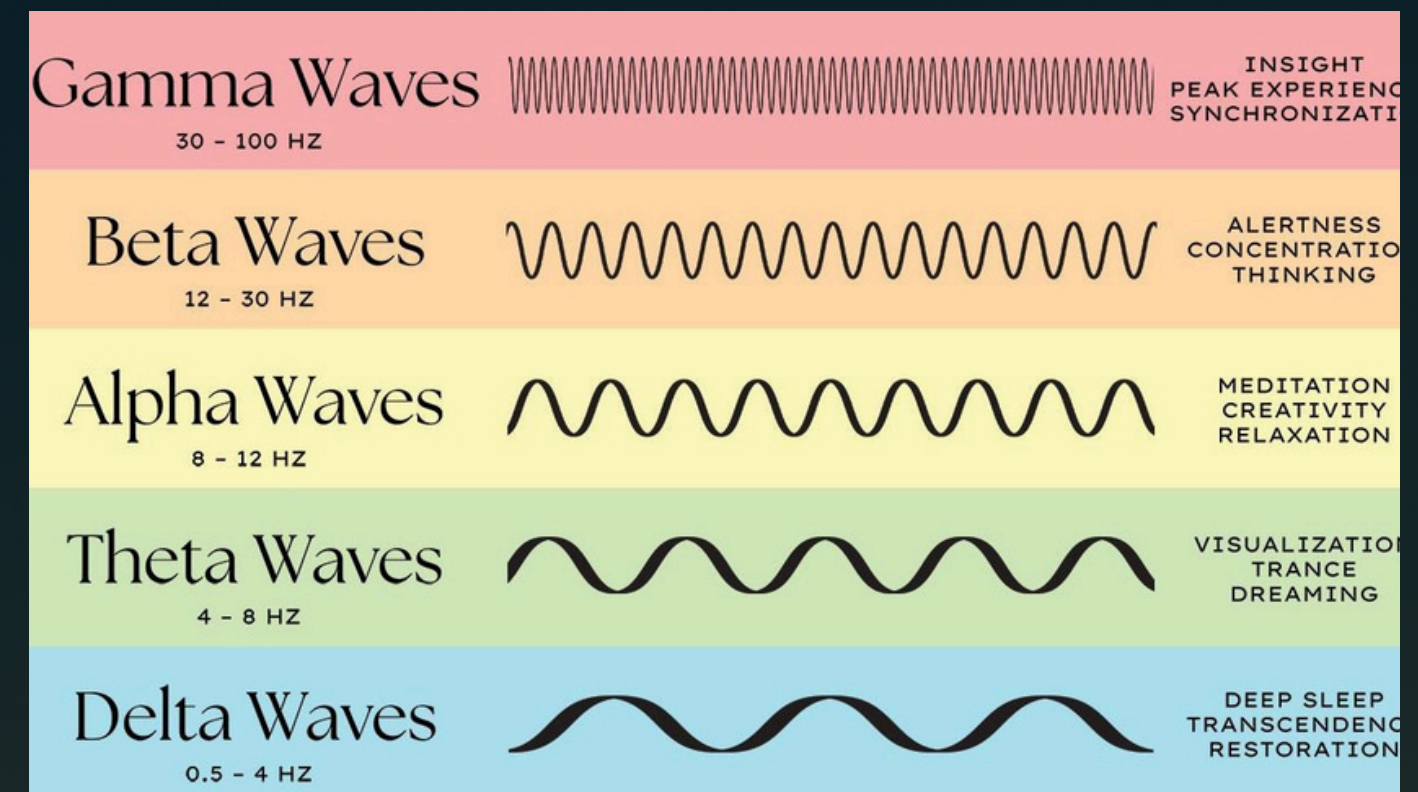
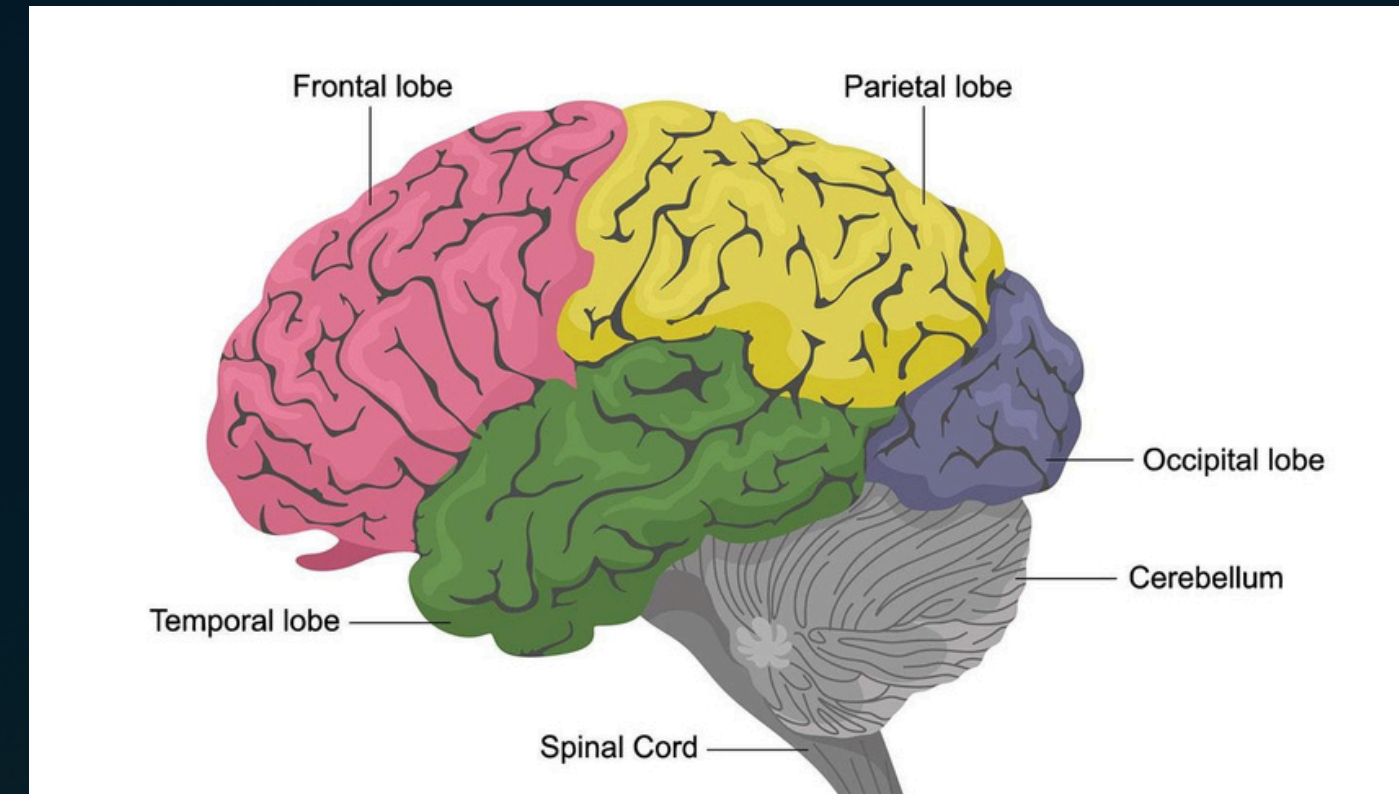


Vs.

Emotiv



- Measures data from all lobes
- Bulky
- 14 electrodes
- 3x more expensive



Materials & Participants

- Hardware
 - Muse 2 headband
 - Exports raw EEG data
 - Macbook Air M3 Chip
- Software
 - Jupyter Notebook
 - For data science / analysis
 - Terminal
 - To set up local environments and kernels
 - Miniconda
 - Manage python environments and packages
- Participants
 - 6 highschoolers (2 female, 4 male, 15 – 17 yo)
 - Healthy with no self reported history of any neurological disorders
 - Conforms to Youth Science Canada ethics regulations (informed consent)



Project: Data Collection

Experimental Setup:

- Participants were placed in a quiet, well-ventilated room to minimize environmental distractions.
- After following the Muse headband placement protocol and ensuring a stable, clean EEG signal, participants were instructed not to touch or adjust the headband, stand up, speak, or engage in unnecessary movement during the recording period in order to reduce motion artifacts and preserve signal quality.
- At this point, the recording starts and the researcher leaves the room.

Experimental Task:

- Participants were instructed to bring any academically demanding work which required sustained attention and engagement to the 30 minute recording session. If participants were unable find any demanding work, they were provided an SAT mock exam instead.
- This level of task intensity was required to induce sustained cognitive load, which accelerates mental fatigue over time and thereby increases the likelihood of observing decreases in neural engagement metrics throughout the relatively short 30 minute session testing window.

Data Collection:

- After 30 minutes, the recording stops and all data is exported to dropbox automatically.
- Next, the recording device is removed and the participant is provided with a self-report questionnaire to complete.
- This form collects subjective measures of perceived focus, mental fatigue, task difficulty, and any discomfort experienced during the session, allowing comparisons between physiological engagement metrics and self-reported cognitive state to be made if desired.



Project: Processing & Filtration

1. Butter worth filter

- Raw EEG data contains a lot of noise.
- This filter processes the raw EEG signal directly. Then, it gradually reduces frequencies above a chosen cutoff threshold, since those higher frequencies are more likely caused by noise or factors unrelated to engagement (Fig. 1).

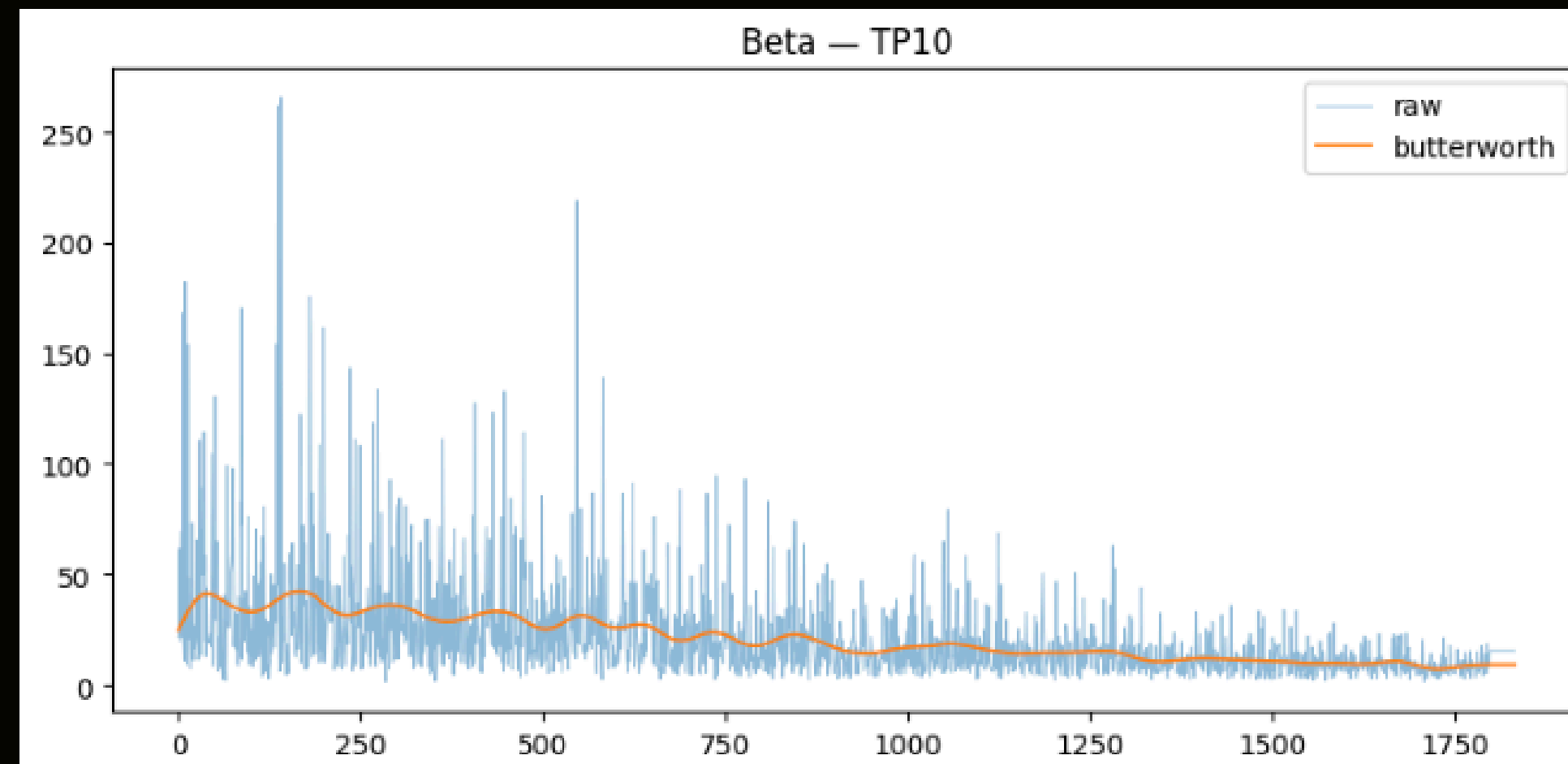


Figure 1. This is the Beta strand of the TP10 electrode. The blue line depicts the raw data whilst the orange line is the result after the butterworth filter has been used.

Project: Processing & Filtration (Cont.)

2. Engagement Index

- a. After we average all the filtered waves, we use the engagement index formula (Fig.2.1): $\text{beta} / (\text{alpha} + \text{theta})$
 - i. There is already established literature associating beta activity with active cognitive processing and alpha/theta activity with reduced attentional engagement.
- b. This calculation produced a continuous engagement time series across the 30-minute task period (Fig.2.2).

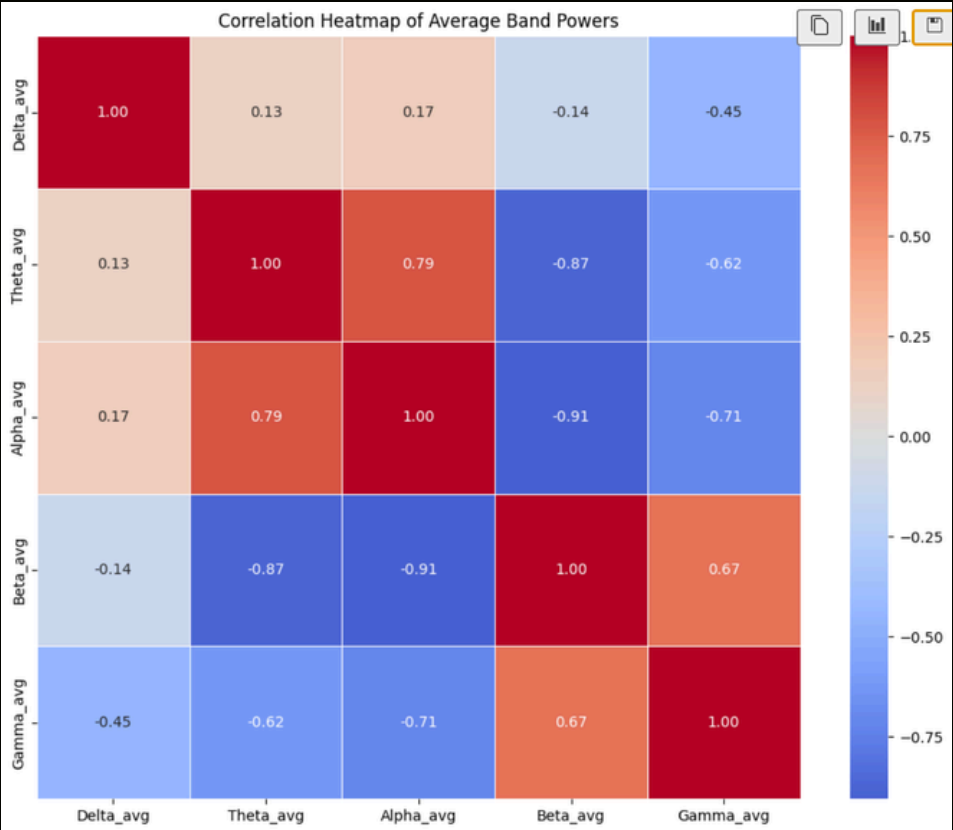


Figure 2.1. This heatmap shows the correlation between different frequency types. Beta & Gamma have strong correlation as well as Alpha & Theta.

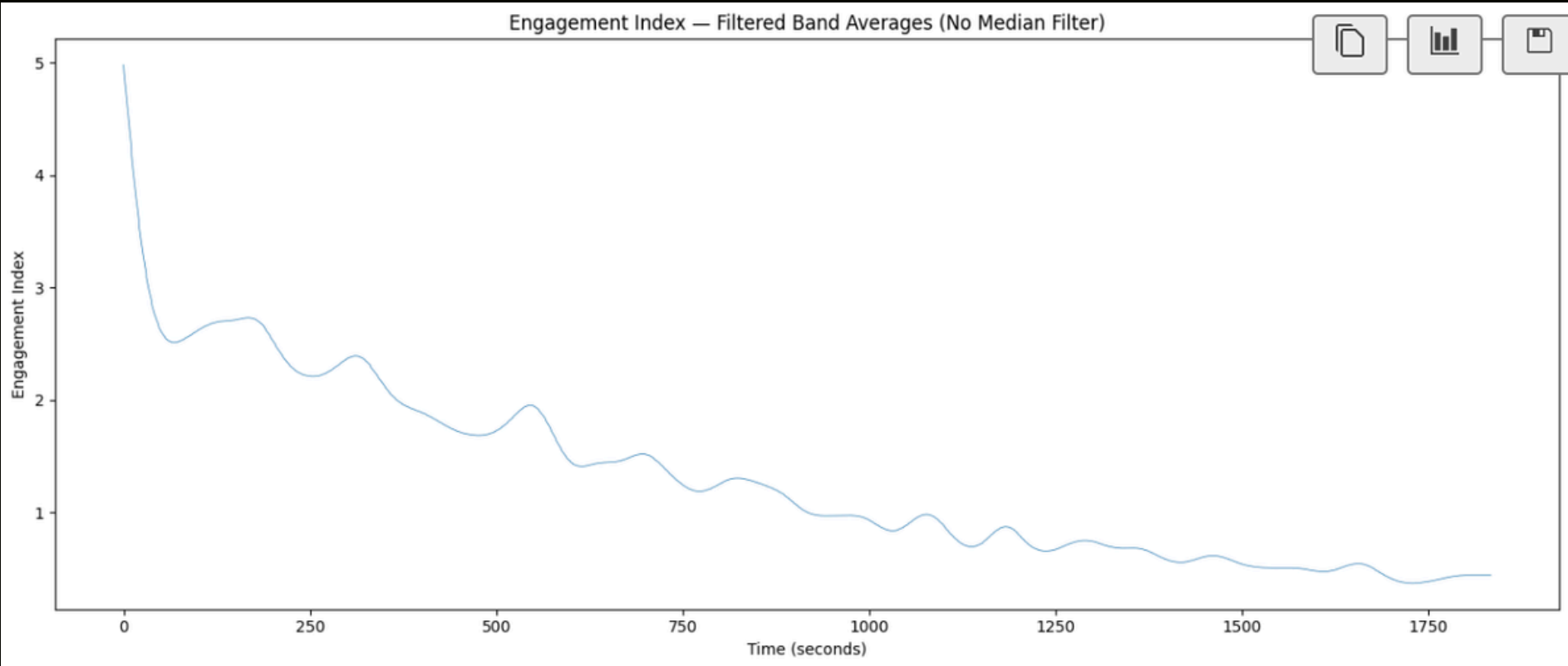


Figure 2.2. Engagement index time series achieved after averaging the aeta, alpha, and theta strands. We apply the EI formula on each frame.

Project: Processing & Filtration (Cont.)

3. Rolling Median

- If there are any extreme outliers remaining, I chose to use a rolling median which preserves the overall trend, whilst removing outliers (Fig.3).
- A rolling median smooths time-series data by computing the median within a fixed window of consecutive points and assigning it to the center value.

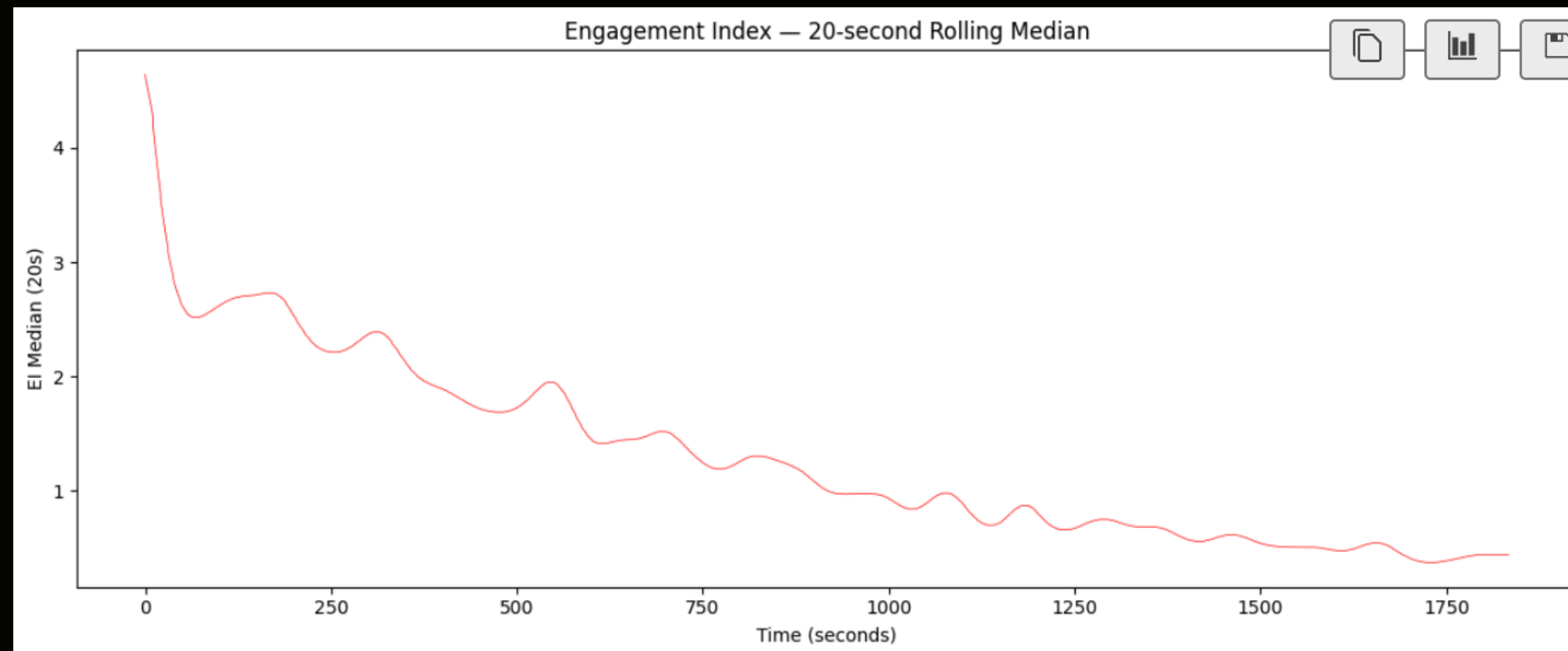


Figure 3. We push the EI time series through a rolling median filter. Severe outliers get omitted and the general trend becomes more prominent.

Project: Product Demo & Limitations

Although we can't stream the EEG signals from the Muse 2 directly to a computer, we can treat the raw CSV as if it was coming live.

In the first 2 - 10 minutes, we calculate the mean and standard deviation (Fig.4).

If the user's EI falls below the threshold, they are determined to be disengaged. If their EI stays below for a sustained period of time, it is determined that they're fatigued.

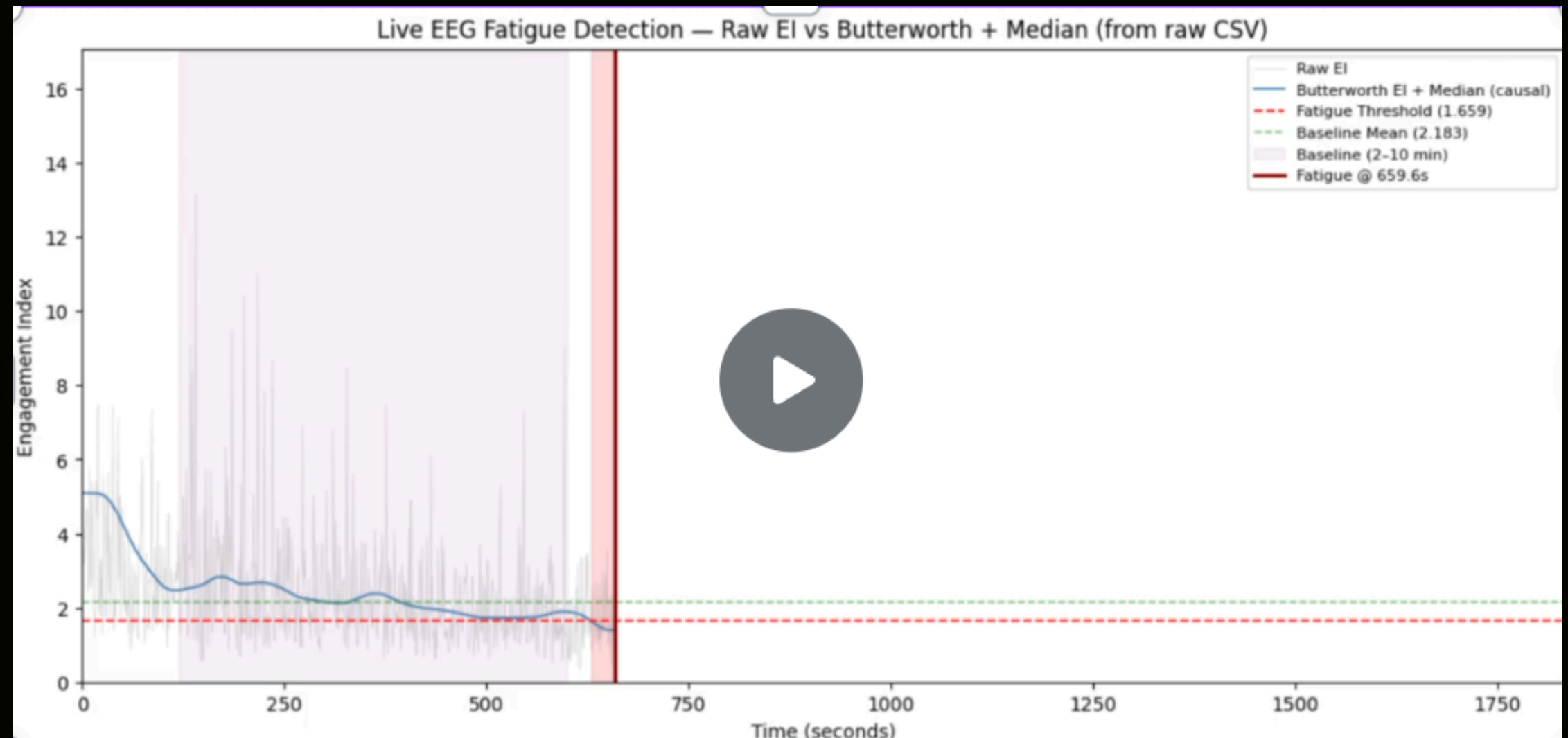


Figure 4. This live demo (sped up 200×) shows real-time engagement tracking. The baseline is calculated from minutes 2–10 (purple region). The green line represents the mean Engagement Index (EI), while the red line marks 1.5 standard deviations below the mean (disengagement threshold). Red-shaded intervals indicate when EI falls below this threshold. If EI remains below the threshold for more than 30 consecutive seconds, the system stops the demo and records the time of disengagement. For the data frame above, that happened at around the 659 second mark.

Applications & Future Steps

Applications

Educational - Currently, this system does not operate in real time; instead, the data is analyzed retrospectively after the session ends. Nevertheless, this still provides valuable insights about individual cognitive fatigue patterns and optimal break timing. Students can use these results to better understand their focus patterns and adjust study strategies, such as modifying break timing or task duration.

Beyond Education - there are potential medical applications. Similar EEG-based monitoring systems could support stroke rehabilitation by reinforcing beneficial neural activity patterns through neuroplasticity. They may also assist in monitoring attention disorders, tracking cognitive fatigue in concussion recovery, or supporting patients undergoing cognitive therapy. In the future, with hardware capable of live EEG streaming, this framework could evolve into a real-time closed-loop system that provides immediate feedback or therapeutic interventions in both academic and clinical settings.

Future Steps

Biomarker Validation - the collected EEG data can be compared with blink rate and heart rate to strengthen the validity of the Engagement Index as a biomarker of cognitive fatigue. By analyzing patterns across multiple participants, this project can contribute to a broader understanding of how cognitive overload develops over time.

Real Time Feedbacks - In the future, with hardware capable of live EEG streaming, this framework could evolve into a true real-time closed-loop system. Such a system could instantly detect declining engagement or rising cognitive load and trigger timely interventions, making focus management more precise and personalized.

Github & Documentation



Ethics

- This study investigates the relationship between sustained academic activity and cognitive engagement, quantified using EEG-derived engagement indices during a 30-minute work session. Conclusions are limited to short-term observations within a controlled experimental setting, and no claims are made regarding long-term cognitive effects.
- The study qualifies under Low Risk Human Participation guidelines as per Youth Science Canada. The EEG device used (Muse headband) is non-invasive and comparable to commercially available wearable technology. Experimental tasks are similar in nature to typical academic activities. Potential risks are minimal and may include temporary mental fatigue or mild physical discomfort from wearing the headband. Participants were informed that they could withdraw at any time without penalty.
- Participation was fully voluntary. The study does not promote or advocate for mandatory cognitive monitoring in educational settings. Any potential future applications of attentional monitoring technologies should prioritize individual autonomy, informed consent, and ethical oversight.
- EEG recordings are treated as sensitive biometric data. Each participant was assigned a coded identifier, and no personally identifying information was attached to the dataset. All data were stored locally on a password-protected device accessible only to the researcher. Only aggregated, anonymized results are reported.
- This research does not diagnose cognitive fatigue, attention disorders, or neurological conditions. Findings are exploratory and correlational in nature, and interpretations are presented conservatively to avoid overstating the capabilities of consumer-grade EEG technology.

References

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